

DataLens Agent Report

Assignment 2: Intelligent Software Agent Prototype

GitHub repo	https://github.com/gdegegiu/DataLens-Agent
Agent type	CSV Data Analysis Agent. The agent can read a CSV dataset, understand its structure, choose suitable analysis steps, run built-in analysis tools, generate charts, and summarize the results in natural language.

1. System Design and Decision Policy

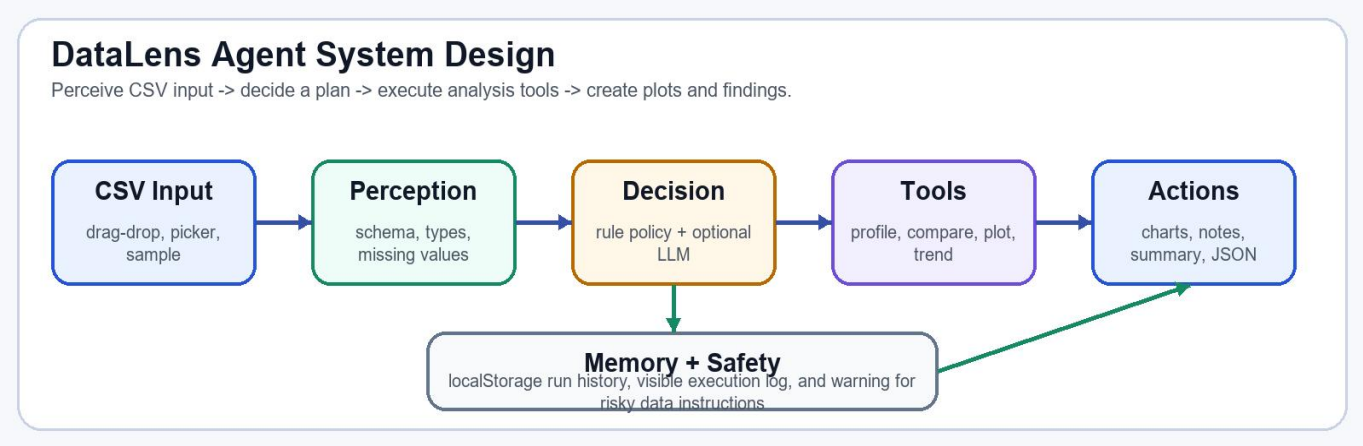


Figure 1. The agent loop combines schema perception, decision making, executable analysis tools, and visible outputs.

Design explanation

DataLens Agent is a CSV Data Analysis Agent. It can read a CSV dataset, understand its structure, choose suitable analysis steps, run built-in analysis tools, generate charts, and summarize the results in natural language.

The system works in a step-by-step agent loop. First, the user uploads, pastes, or selects a sample CSV file. The agent then checks the dataset structure, including rows, columns, numeric fields, category fields, date fields, and missing values. Based on this information, it decides which tools should be used. For example, numeric columns can trigger numeric summaries and histograms, category and numeric columns can trigger group comparison and bar charts, and two numeric columns can trigger scatter plots and correlation analysis.

The agent can also use an optional language model to help create a better analysis plan and explanation. The browser does not contact the model provider directly. Instead, it sends the request to a local backend proxy running on the user’s computer. The proxy then forwards the request to the selected model provider, such as Alibaba Cloud DashScope with the Qwen model . This design is safer for the API key and more flexible because the model provider can be changed without rewriting the front-end code.qwen-plus

After the plan is created, the agent runs the selected tools and shows the results through charts, summaries, execution logs, safety checks, and memory records. If the model service is unavailable or the API key is not provided, the agent can still complete the analysis using deterministic browser-based rules and tools. This makes the prototype reliable, easy to test, and usable even without an external model connection.

Detected condition	Automatic decision
Numeric columns exist	Run numeric profile and offer histogram / scatter plots.
Category + metric exist	Run group comparison and draw a bar chart; pie chart is allowed for small category counts.
Two numeric columns exist	Run relationship plot and compute correlation.
Date + metric or missing data exist	Add trend analysis or missing-value checks when the schema supports them.

Table 1.Detected condition and automatic decision

2. Screenshots and System Behavior

Figures shows a completed sample run: the agent perceived the CSV, selected executable steps with 95% confidence, generated plots and summary, wrote execution logs, and kept memory records.

PERCEPTION

CSV input

+

Upload CSV

Drop CSV here or click to choose
Supports .csv files and pasted CSV text
Selected file: iris.csv

Analysis goal

Analyze this uploaded CSV and summarize the most important patterns.

CSV preview or pasted CSV

sepal_length,sepal_width,petal_length,petal_width,species
5.1,3.5,1.4,0.2,setosa
4.9,3.0,1.4,0.2,setosa
4.7,3.2,1.3,0.2,setosa
4.6,3.1,1.5,0.2,setosa
5.0,3.6,1.4,0.2,setosa
5.4,3.9,1.7,0.4,setosa
4.6,3.4,1.4,0.3,setosa
5.0,3.4,1.5,0.2,setosa
4.4,3.0,1.4,0.3,setosa

☒ Use LLM planner and summary

Model
qwen-plus

API key
.....

LLM calls go through the local /api/llm proxy. The key is stored only in this browser. If the LLM call fails, the agent falls back to deterministic tools.

1.Upload your csv

2.Set a goal and it will show a CSV preview

3.Set LLM

Chosen analysis steps 95% confidence

☒ Agent auto-decision
Let the agent choose analysis steps and chart types from the CSV schema.

☒ Schema and quality

☒ Numeric profile

☒ Group comparison

☒ Relationship plot

☐ Missing values

☐ Outlier scan

☐ Trend analysis

☒ Final summary

Chart types to draw
☒ Bar chart ☒ Scatter plot ☐ Histogram ☐ Pie chart ☐ Line chart

Custom analysis instruction
Optional: ask the agent to check one extra pattern.

4.Choose analysis steps or auto decision

01	Inspect dataset schema	Confirmed column names, types, and basic metadata to guide subsequent analysis. Tool 1: inspect_schema	Executed
02	Generate numeric profiles	Compute descriptive statistics (mean, std, min, max, quartiles) for all numeric columns to understand distributions and scales. Tool 2: numeric_profile	Executed
03	Compare groups by species	Analyze how sepal and petal measurements differ across the three species to identify distinguishing patterns. Tool 3: group_compare	Executed
04	Visualize pairwise relationships	Plot scatterplots (e.g., petal_length vs petal_width) colored by species to reveal clustering and separability. Tool 4: relationship_plot	Executed
05	Scan for outliers	Detect potential outliers in numeric features using IQR or z-score methods, especially given the small sample size. Tool 5: outlier_scan	Executed
06	Summarize key findings	Synthesize insights from prior steps into concise, actionable conclusions about structure, separation, and anomalies. Tool 6: summarize_findings	Executed

EXECUTION

Execution log

1. inspect_schema: Inspect dataset schema

Detected 150 rows, 5 columns, and 0 missing cells.

2. numeric_profile: Generate numeric profiles

Profiled 4 numeric column(s).

3. group_compare: Compare groups by species

Aggregated sepal_length by species and generated the main bar chart.

4. relationship_plot: Visualize pairwise relationships

Created a relationship plot for sepal_length and sepal_width.

5. outlier_scan: Scan for outliers

Scanned 4 numeric column(s) for IQR outliers.

6. summarize_findings: Summarize key findings

Generated final natural-language findings.

7. bin_summary: Generate LLM narrative

LLM rewrote the computed results into final findings.

5.Agent executes the steps and execution log



6.Generate plots

Overall summary

- The dataset contains 150 iris flower measurements across 5 columns (4 numeric, 1 categorical species) with zero missing values. Sepal length averages 5.84 (range 4.3–7.9), sepal width averages 3.06 (range 2.0–4.4), petal length averages 3.76 (range 1.0–6.9), and petal width averages 1.20 (range 0.1–2.5). Virginica has the highest total sepal length (329.4), followed by versicolor (296.8) and setosa (250.3), indicating species-level morphological differences. Sepal length and sepal width show weak negative correlation (-0.12), suggesting minimal linear association. Four potential outliers were detected in sepal width (values at extremes of its range). Histogram of sepal length shows unimodal distribution peaking in the 5.5–6.1 bin (43 instances), supporting overall data consistency. Data quality is high with no missing cells.

7.Gnerate summary

- Automatic analysis:** the agent selected schema, numeric profile, group comparison, relationship plot, outlier scan, and final summary from the sample schema.
- LLM integration:** the UI exposes LLM mode and model/key fields, but the HTML calls only /api/llm; the local proxy forwards to DashScope or another configured provider.
- Fallback reliability:** You can turn LLM mode off; the same tools still completed, the agent is usable even before adds API key.
- Future improvement:** add saved project sessions, larger-file processing, and a visible LLM trace panel that shows model-selected tool calls step by step.